

$$\begin{array}{ccc}
F^T X & \xleftarrow{i_{F^T X}^T} & [(\mathbb{1}, \mathfrak{a}_{\mathbb{1}}), F^T X]_T \\
\uparrow F^T i_X & & \uparrow [\psi_0, F^T X]_T \\
F^T [\mathbb{1}, X] & \xrightarrow{\psi_{\mathbb{1}, X}} & [F^T \mathbb{1}, F^T X]_T
\end{array}$$